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AVION: Aerial Vision–Language Instruction from Offline Teacher to Prompt-Tuned Network

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Abstract

Adapting vision-language models to remote sensing imagery remains challenging due to two key factors: limited semantic coverage in textual representations and insufficient adaptability of visual features. These issues are particularly significant in aerial scenes, which involve various visual appearances and fine-grained object distinctions. We propose AVION, a knowledge distillation framework tailored for remote sensing adaptation of vision-language models. The teacher module constructs semantically rich textual prototypes by collecting descriptions from a large language model and verifying validity using remote sensing image features. The student module integrates lightweight and learnable prompts into both vision and language encoders, guided by the teacher to align embeddings and their cross-modal relationships. Once trained, the student operates independently during inference. Experiments on six optical remote sensing benchmarks show that AVION improves few-shot classification and base-class accuracy without degrading generalization to novel categories. It also enhances mean recall for cross-modal retrieval, with minimal additional trainable parameters. Code is publicly available at https://github.com/yuhu990424/AVION

1. Introduction

Domain-specific vision–language models (VLMs) adapt rich knowledge of general-purpose foundation models to specific applications [2, 15, 35]. For example, RemoteCLIP [25] and GeoRSCLIP [41] have achieved state-of-the-art performance in remote sensing (RS) after extensive training. However, it remains a significant challenge to efficiently adapt these large-scale foundation models to new scenarios. Fully fine-tuning the entire backbone is an ideal option to boost the performance on new tasks, but it is computationally expensive and slow. This approach is often infeasible in real-world scenarios that demand rapid deployment while having only scarce labels and limited computational resources [12, 40].

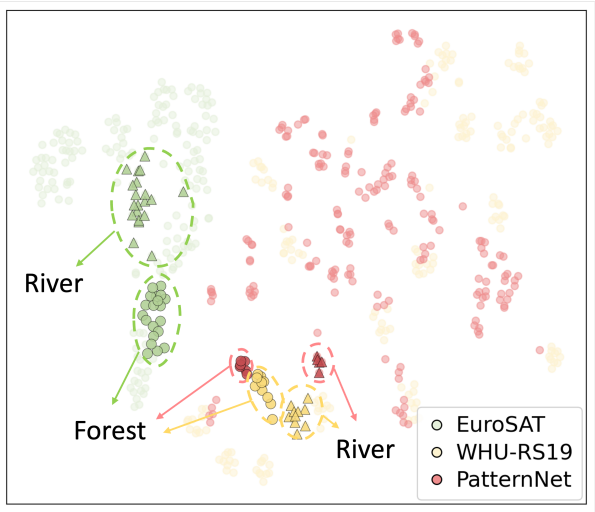


Figure 1. **The t-SNE visualization of visual embeddings from different datasets.** The same class demonstrates large variations across datasets. However, only class names are provided in the datasets, which limits multimodal alignment.

Parameter-efficient fine-tuning (PEFT) emerges as a lightweight adaptation solution [20, 43]. These methods only update a small set of parameters, leaving the backbone of foundation models frozen. While efficient, existing PEFT methods remain sub-optimal for RS. As illustrated in Fig. 1, the same category can have disparate visual appearances across different datasets. However, the images are only paired with the class name, which cannot effectively capture such visual variances. PEFT methods like CoOp [43] learn new tasks from simple templates (*e.g.*, `a`

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`photo of [CLASS]`). As a result, the text encoder cannot adequately describe the diverse appearance patterns in RS data. We denote this limitation as ***semantic poverty***. Such a disconnection between labels and semantics also makes generalization to novel classes challenging.

Moreover, many PEFT methods primarily update the text encoder while freezing the visual encoder. We argue that this design obscures models from capturing scale variation (*e.g.*, large spatial context and subtle textures) and heterogeneity across sources in RS imagery. We refer to this limitation as ***visual rigidity***. Although PEFT can also act on the vision side through visual prompting [19] and lightweight adapters [13], they still struggle to capture RS-specific visual features without proper guidance from the textual side.

To this end, we introduce Aerial Vision–Language InstructiON (**AVION**), a knowledge distillation framework that addresses the above issues. A frozen large teacher model is employed to distill knowledge into a lightweight student model that is later used for inference. To tackle the *semantic poverty*, we use a large language model (LLM) to generate semantically rich descriptions for each class. The teacher model then aggregates these descriptions into a robust textual prototype based on its similarity with the visual embeddings of the corresponding class. This prototype contains much richer information compared with the class name alone. For *visual rigidity*, we inject learnable prompts to the visual side like VPT [19]. These prompts give the visual encoder flexibility to handle visual variations (*e.g.*, oblique views) in the RS scenarios.

Then, we apply distillation on the visual embeddings, textual embeddings, and their similarity logits. This tri-aspect distillation helps the student acquire the visual and textual understandings, as well as the multimodal alignment structure constructed by the teacher model. The learnable prompts for both vision and text encoders allow the student model to accumulate rich knowledge on diverse RS imagery, guided by robust textual prototypes. This flexibility maintains the generalization to unseen data. As shown in Tab. 2, AVION is the only method among state-of-the-art approaches that yields higher base-to-novel accuracy than the GeoRSCLIP baseline.

Our contributions are summarized as follows:

- We identify and analyze two critical limitations of previous PEFT methods for remote sensing imagery: semantic poverty and visual rigidity.
- We propose AVION, a knowledge distillation framework that addresses both limitations with semantically rich textual prototypes and multimodal prompt tuning for both visual and text encoders.
- With a tri-aspect alignment, AVION consistently surpasses other state-of-the-art PEFT methods in few-shot classification, base-to-novel generalization, and frozen-gallery retrieval settings.

2. Related Work

2.1. Vision-Language Models

VLMs align images and text in a shared embedding space and deliver strong zero-shot classification and cross-modal retrieval performances. Contrastive Language-Image Pre-training (CLIP) models train an image and a text encoder in a contrastive fashion [17, 27]. Subsequent advances have refined objectives [30, 39], introduced fine-grained interactions [34], or explored efficient transfer recipes [8, 38]. Other architectures have explored generative/fusion modeling [22, 36] or incorporating LLMs via lightweight bridges (*e.g.*, BLIP-2 [23]).

2.2. Parameter-Efficient Adaptation of VLMs

The heavy foundation models motivate PEFT methods. Among these, prompt learning has emerged as a prominent approach. Text-side prompting (*e.g.*, CoOp [43], CoCoOp [42]) optimizes learnable context tokens, while vision-side prompting (VPT [19]) inserts prompts into ViT layers. Multi-modal prompting (MaPLe [20]) introduces paired prompts for both encoders. Beyond prompting, non-prompt adapters (*e.g.*, CLIP-Adapter [12], Tip-Adapter [40]) attach lightweight heads or training-free caches. MMRL injects a learnable modality-agnostic representation space into higher layers to balance specialization and generalization [13]. PromptKD uses unlabeled images for logit-based distillation from a CLIP teacher to learnable prompts [24]. Prompt-CAM uses PEFT to reveal interpretable traits [9]. While efficient, these PEFT methods are sub-optimal in aligning diverse visual appearances with their class names in the RS scenario. AVION tackles this challenge via introducing semantically rich textual prototypes and prompt tuning for both vision and text encoders.

2.3. VLM Application in Remote Sensing

RS images present a significant visual domain gap compared to general-purpose web data. Unlike ground-level photos, RS images are captured from top-down perspectives, with significant variations in object scale and resolution. The progress of the VLM application has followed two fronts. The first front builds RS-specific foundation models. Works like RemoteCLIP [25], RS5M/GeoRSCLIP [41], and LRSCLIP [6] build large-scale RS datasets to train domain-specific models. Others align images directly with geographic coordinates (*e.g.*, GeoCLIP [3], SatCLIP [21]). The second front focuses on parameter-efficient RS adaptation. For classification, APPLeNet [28] generates image-conditioned visual prompt tokens by fusing multi-scale content features with domain style information. MVP [45] integrates visual prompting into a meta-learning framework. RotCLIP [29] augments CLIP with a lightweight Rot-Adapter and dual textual prompts. RSPrompter [4] and RSRefSeg [5] learn prompts for seg-

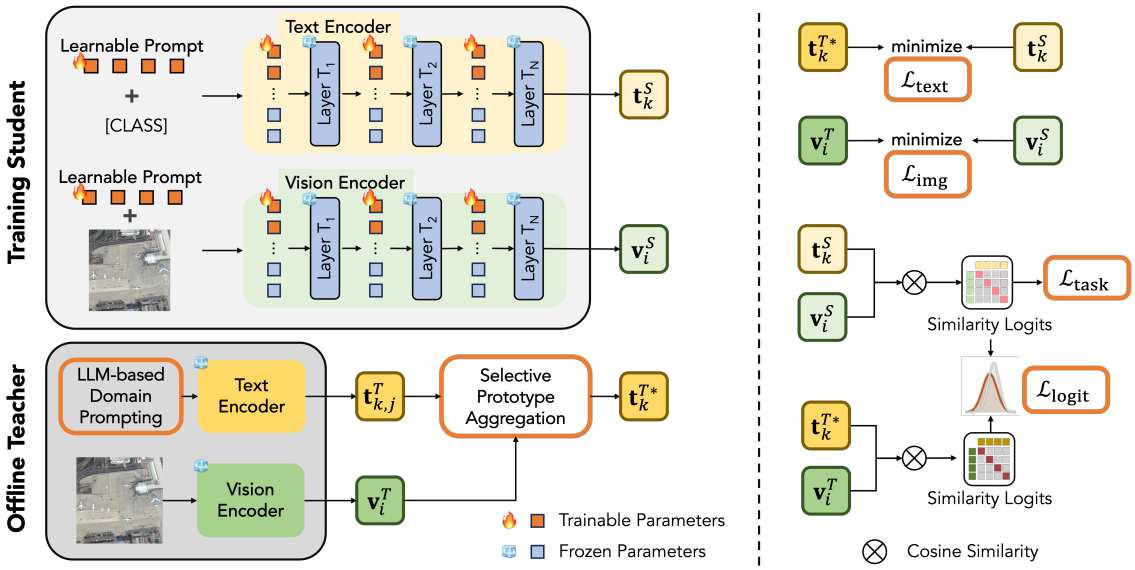


Figure 2. **Overview of AVION. Upper-left (Training Student).** Learnable prompt tokens are injected into the text and vision encoders. The student outputs embeddings: $\mathbf{t}_k^S$ (student text embedding for class k) and $\mathbf{v}_i^S$ (visual embedding for image i). **Bottom-left (Offline Teacher).** *LLM-based Domain Prompting* generates multiple class-aware descriptions, which are encoded into $\mathbf{t}_{k,j}^T$. *Selective Prototype Aggregation* verifies these candidates using teacher visual embeddings $\mathbf{v}_i^T$ and aggregates them into an RS-aware text prototype $\mathbf{t}_k^{T*}$. **Right (Training Objectives).** Tri-aspect alignment: (i) *textual alignment* $\mathcal{L}_{\text{text}}$ maximizes the cosine similarity between $\mathbf{t}_k^S$ and $\mathbf{t}_k^{T*}$; (ii) *visual alignment* $\mathcal{L}_{\text{img}}$ aligns $\mathbf{v}_i^S$ with $\mathbf{v}_i^T$; (iii) *similarity logit alignment* $\mathcal{L}_{\text{logit}}$ matches teacher logits $s_{i,k}^T = (\mathbf{v}_i^T)^\top \mathbf{t}_k^{T*}$ and student logits $s_{i,k}^S = (\mathbf{v}_i^S)^\top \mathbf{t}_k^S$ via temperature-scaled KL. A standard task loss $\mathcal{L}_{\text{task}}$ (cross-entropy) is applied on the student.

mentation. PeftCD [11] plugs LoRA and Adapter modules into Siamese backbones for change detection. Compared with other explorations, AVION focuses on the intrinsic visual variation of RS imagery. The distillation improves adaptation results under realistic settings while retaining efficiency.

3. Methodology

Compared with general web images, remote sensing (RS) imagery poses unique challenges [10, 16]. First, RS images present a large gap from general images, which stems from the top-down perspective. Second, within RS domains, the same semantic category can have drastically different visual appearances due to the variations of regions, seasons, sensor types, and resolutions. These challenges make existing PEFT methods inadequate for adapting foundation models to new scenarios. In this work, we propose an AVION distillation framework for more robust adaptation performance without sacrificing efficiency. An overview of the proposed AVION framework is illustrated in Fig. 2.

3.1. Preliminary

Let $\mathcal{X}=\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ be the training set, where the label $y_i \in \mathcal{Y}=\{1,\dots,C\}$. We employ a large teacher model in the training process to provide distillation supervision. Learnable deep prompts are injected into a student model whose backbone parameters are frozen [18]. We denote the visual and textual embeddings encoded by the teacher model as $\mathbf{v}^T$ and $\mathbf{t}^T$, respectively. Similarly, the student's visual and textual embeddings are denoted as $\mathbf{v}^S$ and $\mathbf{t}^S$. Logit $s_{i,k}$ represents the similarity between visual embedding $\mathbf{v}_i$ and textual embedding $\mathbf{t}_k$. We present a summary of all the symbols and hyperparameter settings in Appendix B.

As discussed, adapting VLMs to RS is challenged by two critical limitations: (1) *semantic poverty*, as simple class names fail to capture fine-grained visual distinctions; (2) *visual rigidity*, as the frozen visual encoder struggles with RS-specific features. There is a necessity for a framework that enriches textual supervision while enhancing visual adaptability to mitigate the gap of RS-tailored adaptation.

Therefore, we propose to design a distillation framework that efficiently adapts RS foundation models to new application scenarios. Specifically, we propose to enhance the textual prototype to incorporate richer semantic information as distillation supervision. Moreover, the proposed AVION framework aligns both embeddings and similarity logits during distillation.

3.2. Textual Prototype Enhancement

The teacher model encodes images and text labels as distillation supervisions for the student model. Compared with the visual embeddings conditioned on different images, the adopted RS datasets often only provide the class name without detailed descriptions. Therefore, the textual embeddings

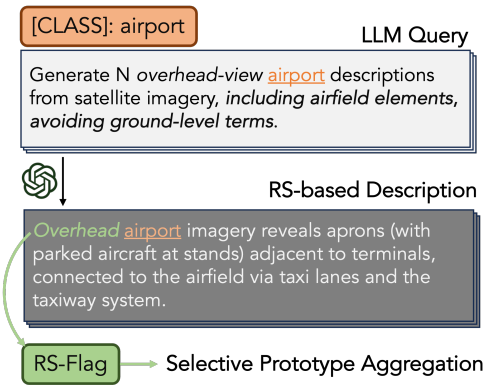


Figure 3. **LLM-based Domain Prompting.** Given a class k, an RS-aware query asks the LLM to produce aerial-view descriptions. `RS-Flag` is used to examine whether the description contains RS-related tokens.

can be regarded as class-specific prototypes. The corresponding visual embeddings are optimized toward the textual prototype during distillation. It also indicates that for a dataset with C classes, it will also have C different textual prototypes. As aforementioned, the class names alone cannot provide sufficient information for the intrinsic visual variations across different RS scenarios. Thus, we propose incorporating rich knowledge of large language models (LLMs) to generate descriptions for RS classes.

LLM-based domain prompting. Given a class $k \in \mathcal{Y}$, we prompt LLM to generate the corresponding descriptions in the context of RS. As shown in Fig. 3, the generated descriptions largely improve the semantic richness of the textual supervision. The detailed prompt and concrete examples of these generated candidates are presented in Appendix C.

While presenting rich semantic information, LLMs may hallucinate non-visual or non-RS-related descriptions. The textual information might not be grounded in the images as well. Therefore, it is critical to verify the validity of these descriptions based on images in each dataset. We attach an `RS-Flag` to each description based on the following rules: the description should contain RS-related words and exclude RS-negative words; the description should be within a proper length; the description should include class-specific cues. Details of `RS-Flag` are in Appendix D.

Selective prototype aggregation. To verify the LLM candidates, we first cache the teacher visual embeddings $\mathbf{v}_{k,i}^T$ of class k and obtain the visual prototype $\widehat{\mathbf{v}}_k^T$ by:

$$\widehat{\mathbf{v}}_k^T = \frac{1}{|\mathcal{B}_k|} \sum_{i \in \mathcal{B}_k} \mathbf{v}_{k,i}^T, \tag{1}$$

where $\mathcal{B}_k$ is the sample set of the class k.

LLM candidates may include ground-view phrases or off-topic wording. As shown in Fig. 4, visual prototypes are used to *score* these candidates. We calculate the similarity

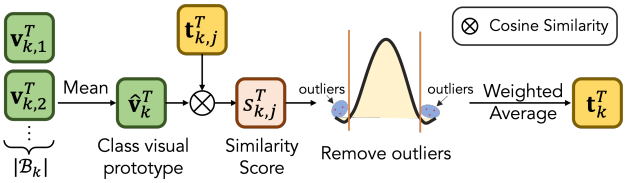


Figure 4. **Selective Prototype Aggregation.** For each class k, teacher image embeddings $\{\mathbf{v}_{k,i}^T\}_{i \in \mathcal{B}_k}$ are averaged to form the visual prototype $\widehat{\mathbf{v}}_k^T$. Each LLM-generated description is encoded to a teacher text embedding $\mathbf{t}_{k,j}$ and scored by cosine similarity $s_{k,j} = (\widehat{\mathbf{v}}_k^T)^\top \mathbf{t}_{k,j}^T$. A median/MAD threshold $\pm\zeta_s$ removes outlier candidates. The remaining embeddings are combined with softmax-normalized weights (over kept j) to obtain the class textual prototype $\mathbf{t}_k^{T*}$.

between the textual embedding $\mathbf{t}_{k,j}^T$ of each candidate description for class k and the corresponding visual prototype:

$$s_{k,j} = (\widehat{\mathbf{v}}_k^T)^\top \mathbf{t}_{k,j}^T. \tag{2}$$

Next, we remove outlying or generic candidates using a median/MAD-based robust z-score (closed form in Appendix D, Eq. 9). We denote the kept index set by $\mathcal{J}_k$. The kept candidates are then aggregated based on the `RS-Flag` acquired from the last step to obtain the final prototype $\mathbf{t}_k^{T*}$:

$$w_{k,j} \propto \exp\big(\beta s_{k,j} + \gamma\, \texttt{RS-Flag}_{k,j}\big), \quad j \in \mathcal{J}_k, \tag{3}$$

$$\mathbf{t}_k^{T*} = \text{norm}\Big(\sum_{j \in \mathcal{J}_k} \frac{w_{k,j}}{\sum_{j' \in \mathcal{J}_k} w_{k,j'}} \mathbf{t}_{k,j} \Big), \tag{4}$$

where $\texttt{RS-Flag}_{k,j} \in \{0, 1\}$.

This aggregation process is analogous to a parameter-free cross-attention mechanism. The class visual prototype ($\widehat{\mathbf{v}}_k^T$) acts as the *query*, and the candidate text embeddings ($\mathbf{t}_{k,j}^T$) serve as both *keys* and *values*. The `RS-Flag` prior is then added as a *calibration* to the scores before the softmax-like normalization (Eq. 3). Then, the weights $w_{k,j}$ are normalized over the kept set $\mathcal{J}_k$. In such a way, we obtain the textual prototype $\mathbf{t}_k^{T*}$ for class k generated by the teacher model. This prototype contains richer semantic information compared with solely the class name, and also corresponds to the visual features in the target adaptation scenario.

3.3. Tri-Aspect Alignment

After acquiring the visual and textual embeddings of the teacher model, as illustrated in Fig. 2, we apply distillation training to update the learnable prompts in the student model. First, we apply the task supervision $\mathcal{L}_{\text{task}}$ (cross-entropy in this task) to ensure the student model achieves the desired task performance. Additionally, we apply three complementary alignment losses to ensure comprehensive knowledge transfer:

- **Visual alignment ($\mathcal{L}_{\text{img}}$):** We encourage the student's visual embeddings $\mathbf{v}_i^S$ to mimic the teacher's $\mathbf{v}_i^T$:

$$\mathcal{L}_{\text{img}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \left(1 - (\mathbf{v}_i^S)^\top \mathbf{v}_i^T\right), \tag{5}$$

  where $\mathcal{B}$ is the current mini-batch. By injecting learnable prompts into the vision side, we aim to address the *visual rigidity* of previous methods. This objective guides the student to "see" RS-specific geometric patterns (*e.g.*, oblique views) as the teacher model does.
- **Textual alignment ($\mathcal{L}_{\text{text}}$):** This loss provides the primary supervision signal for the student's learnable text prompts. Its effect is to align the student's text embedding ($\mathbf{t}_k^S$) with the high-quality, visually-verified teacher prototypes ($\mathbf{t}_k^{T*}$):

$$\mathcal{L}_{\text{text}} = \frac{1}{C} \sum_{k=1}^{C} \left(1 - (\mathbf{t}_k^S)^\top \mathbf{t}_k^{T*}\right). \tag{6}$$

  By minimizing the cosine distance, the student can produce semantic representations that mimic the rich, LLM-guided knowledge captured by the teacher, thereby combating *semantic poverty*.
- **Similarity logit alignment ($\mathcal{L}_{\text{logit}}$):** This loss aligns the student's predictive output distribution with the teacher's (Eq. 7). In contrast to $\mathcal{L}_{\text{task}}$, which uses a one-hot vector representing only the single ground-truth class, this objective provides a more informative supervisory signal. The effect is that the student is trained to replicate the teacher's *entire probability distribution* over all classes:

$$\mathcal{L}_{\text{logit}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \tau^2 \, \text{KL}\Big(\sigma(\mathbf{s}_{i,\cdot}^{(T)}/\tau) \parallel \sigma(\mathbf{s}_{i,\cdot}^{(S)}/\tau)\Big). \tag{7}$$

  This distribution contains rich, implicit information about inter-class relationships, capturing subtle similarities that are lost in the one-hot target.

The overall objective is a weighted sum:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{img}}\mathcal{L}_{\text{img}} + \lambda_{\text{text}}\mathcal{L}_{\text{text}} + \lambda_{\text{logit}}\mathcal{L}_{\text{logit}}. \tag{8}$$

The teacher is only used during training. We present the complexity analysis in Appendix F and detailed hyper-parameter settings in Appendix B.

4. Experiments

4.1. Experimental Setup

We evaluate AVION on standard RS benchmarks for few-shot classification, base-to-novel generalization, and cross-modal retrieval.

Few-shot learning. To assess the performance of AVION under limited supervision, which is common in remote sensing, we evaluate few-shot classification under different settings of labeled examples per class ($K \in \{1, 2, 4, 8, 16\}$). To ensure a fair comparison, we fix identical per-class samples across all runs for each method and K.

Base-to-novel generalization. Each dataset is split into disjoint base/novel classes. Models are trained on base classes with 16-shot supervision and evaluated on both sets. We report the harmonic mean (HM) following prompt-learning practice [13, 20, 42, 43]. To prevent information leakage from novel classes, the teacher builds prototypes only for base classes. Please refer to Appendix E for the masking mechanism.

Cross-modal retrieval. To perform cross-modal retrieval, we extract image and text representations. These are ℓ_2-normalized, and we retrieve the most similar samples using cosine similarity (the normalized dot product). We report retrieval recall at R@1, R@5, and R@10 for both image-to-text (I→T) and text-to-image (T→I) directions. We also report the mean recall (mR), the average of all six values.

Datasets. For classification (few-shot and base-to-novel), we use AID [32], RESISC-45 [7], EuroSAT [14], WHU-RS19 [31], PatternNet [44], and UCMerced [33], covering diverse spatial resolutions, class counts (19–45), and scene complexities. For cross-modal retrieval, we use RSITMD [37] and RSICD [26].

Implementation details. Unless noted, the student backbone is `GeoRSCLIP(ViT-B/32)` [41] and the frozen teacher is `GeoRSCLIP(ViT-H/14)` [41]. All teacher-side visual prototypes and textual prototypes generated by selective prototype aggregation are prepared offline and used only during training. Specifically, we employ the Gemini 2.5 Flash API for LLM-based domain prompting to generate up to 50 class-specific descriptions per class. For training, we use AdamW with a learning rate of 5×10^{-4} and a batch size of 4. Few-shot runs train for 100 epochs; base-to-novel and retrieval for 50 epochs. For the distillation objective, we set the final weights to λ_{img}=0.5, λ_{text}=0.5, and λ_{logit}=1.0, using a 30% linear warm-up for the λ_{logit} term to ensure stable convergence. The distillation temperature τ is fixed at 2. All hyperparameters were selected on a validation split of the AID dataset and then fixed for all other datasets and tasks. We provide a detailed justification for these choices, including sensitivity analysis, in Appendix G. All experiments were run on a single NVIDIA L4 GPU (24 GB).

4.2. Few-shot Evaluation

We compare AVION with two groups of baselines. (1) Zero-shot VLMs: CLIP [27], RemoteCLIP [25], and GeoRSCLIP [41], all evaluated in the zero-shot mode. (2) Parameter-efficient fine-tuning (PEFT): (a) prompt-learning methods CoOp [43], CoCoOp [42], MaPLe [20], PromptKD [24], and APPLeNet [28]; (b) a representation-token based method MMRL [13]. To ensure fairness, all PEFT

Table 1. Few-shot classification against state-of-the-art methods. The table presents the average classification accuracy (%) over 6 datasets. The best-performing result for each K-shot setting is in **bold**, and the runner-up is underlined.

Method	1-shot	2-shot	4-shot	8-shot	16-shot
Zero-shot Methods					
CLIP [27]			63.24		
RemoteCLIP [25]			72.92		
GeoRSCLIP [41]			**72.95**		
PEFT Methods					
CoOp [43]	69.98	78.95	84.52	87.57	90.24
CoCoOp [42]	70.27	80.56	85.74	88.93	91.41
MMRL [13]	70.57	79.47	84.66	87.48	90.45
MaPLe [20]	65.60	76.69	82.48	88.33	90.73
PromptKD [24]	69.91	79.64	85.02	88.27	90.86
APPLeNet [28]	74.27	81.79	86.40	89.20	91.61
AVION (Ours)	**74.27**	**81.86**	**88.31**	**91.85**	**93.69**

Table 2. Average base-to-novel results (%) across all 6 datasets (ViT-B/32 student). We report Base, Novel, and HM.

Method	Base	Novel	HM
GeoRSCLIP [41]	78.10	79.75	78.81
CoOp [43]	91.17	69.52	78.88
CoCoOp [42]	91.85	70.69	80.06
MMRL [13]	93.75	73.69	82.76
MaPLe [20]	92.87	74.66	82.90
PromptKD [24]	93.67	76.58	84.20
APPLeNet [28]	93.84	75.75	83.84
AVION (Ours)	**95.64**	**79.94**	**87.05**

baselines are implemented on the same CLIP-like backbone GeoRSCLIP, keeping the image/text encoders frozen while training only method-specific lightweight parameters.

As shown in Tab. 1, AVION matches or exceeds other methods across all shot levels. At 1-shot, it ties the top method (74.27%), and its performance margin over the runner-up progressively widens with more training samples, peaking at +2.65 pp at 8-shot. We attribute this high data efficiency to learnable visual prompts and semantically rich textual prototypes generated by the teacher. The textual prototypes are then effectively incorporated via our tri-aspect alignment objective, capturing more supervision signals. Even if the adopted initialization (GeoRSCLIP) already achieves strong zero-shot performance [41], with RS-specific optimizations in both teacher and student models, our method yields even more significant improvements.

4.3. Base-to-Novel Generalization

The base-to-novel setting evaluates the adaptation to base classes and the generalization to novel classes after fine-tuning. As shown in Tab. 2, with a strong initialization (GeoRSCLIP), all previous methods achieve higher performances on base classes. However, the adaptation also

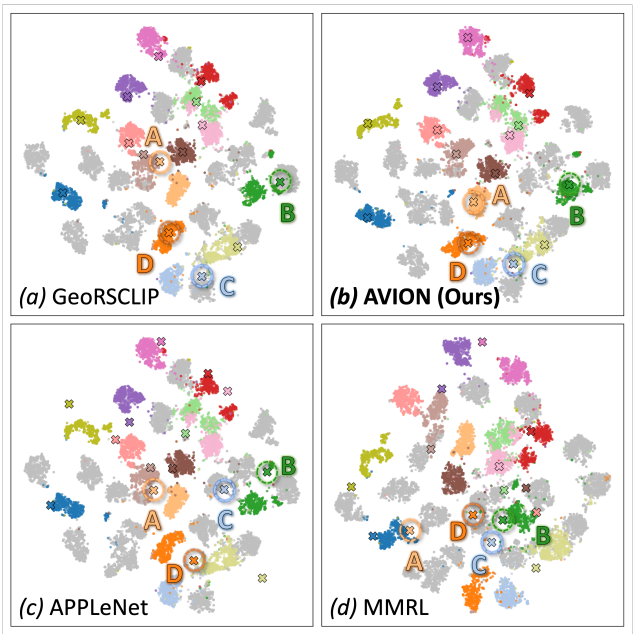


Figure 5. **t-SNE of base-to-novel visual and text embeddings on RESISC-45.** (a) GeoRSCLIP [3] (b) **AVION (ours)** (c) APPLeNet [28] (d) MMRL [13]. Gray dots represent the visual embedding of *base classes*. Colored *dots* represent the *visual embedding of novel classes* with each color standing for a class. Colored *crosses* represent the *text embedding of novel classes* for their class names. A, B, C, and D are four classes.

leads to degraded performance on novel classes. In our comparison, AVION is the only method that surpasses GeoRSCLIP [41] in novel-class performance. Meanwhile, AVION also achieves the best performance on base classes, surpassing the runner-up by +1.80pp.

We present the t-SNE visualization of visual and text embeddings for both base and novel classes in Fig. 5. Gray clusters correspond to base classes, while colored clusters denote novel classes (each color indicates a different class). Colored crosses represent the text embedding of class names for the corresponding colored classes. A, B, C, and D are four representative classes. We highlight their text embeddings with dashed circles. Overall, AVION shows slightly better (*e.g.*, classes A and B) or on-par multimodal alignment on novel classes compared with GeoRSCLIP [41]. It corresponds to the quantitative results in Tab. 2. By contrast, the other two methods, APPLeNet [28] and MMRL [13], illustrate misalignment between text and visual embeddings. APPLeNet freezes the visual encoder and updates only the text encoder. While the visual embedding distribution remains consistent with GeoRSCLIP, the text embeddings on these unseen classes shift after tuning. MMRL opens both encoders for training, but without semantically rich text prototypes as guidance, the multimodal gap becomes even larger for unseen classes.

Table 3. Cross-modal retrieval (**R@K**, %) on RSITMD [37] and RSICD [26] (ViT-B/32 student; protocol details in Sec. 4.1). **mR** is the macro-average over $\{\mathrm{T} \rightarrow \mathrm{I}, \mathrm{I} \rightarrow \mathrm{T}\} \times \{1, 5, 10\}$. Δ is relative to GeoRSCLIP. Our method (AVION) is lightweight, updating only prompts/heads; see Appendix F for a full efficiency analysis.

Testing Dataset	Training Dataset	Method	Image → Text			Text → Image			mR
			R@1	R@5	R@10	R@1	R@5	R@10	
RSITMD [37]	RSITMD	ML Retrieval [1]	19.09	40.26	54.42	17.61	49.73	60.59	41.38
	RET-3	CLIP [27]	23.67	50.00	63.27	22.61	55.27	69.87	47.63
	RET-3 + DET-10 + SEG-4	RemoteCLIP [25]	27.88	50.66	65.71	22.17	56.46	67.41	49.38
	RSITMD + RSICD	GeoRSCLIP-FT [41]	32.30	53.32	67.92	25.04	57.88	74.38	51.81
	RSITMD + RSICD	**AVION (Ours)**	**33.17**	**54.46**	**69.18**	**26.09**	**59.22**	**75.41**	**52.92**
RSICD [26]	RSICD	ML Retrieval [1]	10.70	29.64	41.53	9.14	29.98	44.59	27.43
	RET-3	CLIP [27]	17.94	35.94	51.04	13.89	35.15	50.08	33.98
	RET-3 + DET-10 + SEG-4	RemoteCLIP [25]	17.02	39.97	51.51	13.73	34.71	48.28	35.26
	RSITMD + RSICD	GeoRSCLIP-FT [41]	21.13	41.72	55.63	15.59	41.19	57.99	38.87
	RSITMD + RSICD	**AVION (Ours)**	**22.01**	**42.37**	**56.42**	**16.24**	**42.58**	**59.21**	**39.80**
		Δ vs. GeoRSCLIP-FT (mR)			**RSITMD:** +1.11			**RSICD:** +0.93	

Table 4. **Configurations for our incremental ablation study.** Each configuration builds upon the previous one and culminates in the full system (B7).

ID	Configuration Description	ID	Configuration Description
B0	Shallow text-only prompt (CoOp-style), $\mathcal{L}_{\text{task}}$ only.	B4	Replace manual prototypes with LLM-generated ones.
B1	+ Deep prompt tuning.	B5	+ *Selective prototype aggregation* (vision-guided).
B2	+ Representation alignment ($\mathcal{L}_{\text{img}}$).	B6	+ Behavioral alignment ($\mathcal{L}_{\text{logit}}$) in a single-stage schedule.
B3	+ Semantic alignment ($\mathcal{L}_{\text{text}}$) with manual textual prototypes.	**B7**	**Final Model**: 30% warm-up (Sec. 4.1).

The teacher textual prototypes in AVION integrate rich knowledge from LLMs into the distillation process. Through verification with dataset-specific visual prototypes, AVION accurately aligns various visual appearances with their corresponding labels and descriptions. The knowledge, in addition to class names, facilitates a comprehensive understanding of base classes, which further generalizes to novel classes.

4.4. Cross-modal Retrieval

We evaluate cross-modal retrieval using a deployment-oriented protocol. To simulate scenarios where the gallery is fixed, gallery embeddings are pre-computed and are not re-encoded during adaptation. Concretely, AVION uses a lightweight prompt learning approach, which requires updating only a minimal set of parameters (well below 1% of the backbone) and no gallery recomputation. This resource-efficient design is detailed in Appendix F.

As shown in Tab. 3, AVION attains the best mR across both directions with lightweight adaptation on RSITMD and RSICD. Compared with GeoRSCLIP-FT, which opens all the parameters for tuning, AVION improves mR by +1.11 pp (RSITMD) and +0.93 pp (RSICD). This indicates the computational efficiency of our framework. When the trainable parameters are fewer, we introduce additional rich knowledge from LLMs to promote the understanding of these RS classes. The results demonstrate that this semantically richer supervision and our tri-aspect alignment objective are more effective than naive full fine-tuning.

4.5. Ablation Study

Incremental ablation analysis. We conduct an incremental ablation study to quantify the contribution of each component of AVION. Starting from a minimal prompt-only baseline, we enable deep prompt tuning, textual prototype enhancement, and the tri-aspect alignment objective ($\mathcal{L}_{\text{img}}, \mathcal{L}_{\text{text}}, \mathcal{L}_{\text{logit}}$). Eight configurations (B0–B7) and their results are listed in Tab. 4 and Tab. 5, respectively. All runs share identical training budgets, datasets, and evaluation protocols. To avoid leakage in the base-to-novel protocol, the offline teacher is used only during training, and its guidance (textual prototypes and distillation targets) is restricted to base classes with masked renormalization. More details are presented in Appendix E.

Injecting deep prompts (B1) alone improves the base-class accuracy, but also leads to a -16.01 pp degradation on novel classes. When gradually adding more training samples, the improvement of deep prompts becomes marginal, indicating unstable effects.

Adding visual alignment (B2) acts as an effective reg-

Table 5. **Main ablation results for classification.** We report Base, Novel, and Harmonic Mean (HM) accuracy. Metrics are averaged over six datasets.

ID	Base	Novel	HM	K-shot				
				1	2	4	8	16
B0	91.17	69.52	78.88	69.98	78.95	84.52	87.57	90.24
B1	93.52	53.51	66.71	66.95	84.68	84.61	87.34	91.25
B2	89.74	61.15	72.74	70.21	82.99	85.31	89.65	91.51
B3	91.21	68.63	78.34	71.98	83.72	85.90	90.43	92.18
B4	91.03	74.08	81.67	72.15	84.71	87.43	91.89	92.14
B5	91.71	75.87	83.05	72.52	85.06	87.91	92.53	92.65
B6	94.82	79.14	86.26	73.71	**85.34**	**89.28**	**92.95**	93.17
B7	**95.64**	**79.94**	**87.05**	**74.27**	81.86	88.31	91.85	**93.69**

ularizer, partially reversing the generalization degradation (HM +6.03 pp). Except for the 2-shot case, B2 outperforms B0 across all few-shot configurations. This supports the necessity of introducing the visual alignment $\mathcal{L}_{\text{img}}$.

Introducing our enhanced textual prototypes at this stage delivers the largest cumulative HM improvement (+10.31 pp from B2 to B5). This trajectory is paralleled in the few-shot domain, which sees cumulative gains from B2 to B5 of +2.31 pp (1-shot) and +1.14 pp (16-shot). These results are consistent with the hypothesis that high-quality, visually grounded semantics are pivotal for both generalization and data efficiency.

Finally, adding similarity logit alignment ($\mathcal{L}_{\text{logit}}$, B6) and an optimized warm-up schedule (B7) yields a further HM gain (+4.00 pp from B5 to B7). This final optimization pushes HM, 1-shot, and 16-shot to their peak values. Overall, B7 provides the best overall performance, particularly in generalization (HM) and extreme few-shot scenarios.

Ablating the teacher text-prototype pipeline. We fix the model architecture and training objectives for all settings, varying only prototype sources and aggregation in Tab. 6. Handcrafted multi-templates (P1) serve as a baseline (79.91 HM). Switching to domain-constrained LLM candidates with simple averaging (P3) already improves this to 81.70 HM. This highlights the importance of the additional knowledge introduced by LLMs. *Selective prototype aggregation* (P6) further boosts the score to 83.05 HM. This +1.35 pp gain (P6 vs. P3) isolates the benefit of our aggregation over a simple mean. The value of its *visual* guidance is also supported by P5 outperforming a text-only selector (P4, 82.05 HM). We also demonstrate the result of selecting a random subset from candidates (P5), which results in a score of 80.31 HM. Notably, random selection performs even worse than averaging all unconstrained candidates (P2, 80.85 HM). The contrast further supports that P6's gains stem from its vision-guided selection criteria.

Backbone variants (CLIP family). We compare AVION's performance using three different ViT-B/32 backbones:

Table 6. **Teacher text-prototype ablation (P0–P6).** All settings use the same objectives, with different prototype sources and aggregation. HM averaged over the same datasets/seeds.

ID	Prototype Source	Aggregation	`RS-Flag`	HM (%)
P0	Manual (single template)	none	–	79.52
P1	Manual (multi-templates)	mean	–	79.91
P2	LLM (unconstrained)	mean	no	80.85
P3	LLM (domain-constrained)	mean	yes	81.70
P4	LLM (domain-constrained)	text-only selection	yes	82.05
P5	LLM (random subset)	mean	no	80.31
P6	**LLM (domain-constrained)**	***Selective agg.***	**yes**	**83.05**



Figure 6. Few-shot classification accuracy of AVION with different CLIP-based backbones in the few-shot setting.

CLIP, RemoteCLIP, and GeoRSCLIP. The choice of backbone has a clear impact. As seen in Fig. 6, the domain-pretrained GeoRSCLIP achieves the highest accuracy across all settings, from zero-shot up to 16-shot. RemoteCLIP, which is also pre-trained for remote sensing, is the clear second-best. The general CLIP backbone performs last in all cases. This result aligns with the expectation that backbones already familiar with aerial imagery provide a stronger starting point for adaptation. We also observe that, starting from 1-shot, the performance for all variants consistently improves over the zero-shot initializations.

5. Conclusion

We addressed two challenges in adapting vision–language models to remote sensing imagery: semantic poverty in text representations and visual rigidity in feature adaptation. We proposed AVION, a knowledge distillation framework in which a frozen teacher constructs semantically rich textual prototypes from LLM-generated descriptions and remote sensing image features, and a lightweight student learns multimodal prompts for both vision and language encoders under a tri-aspect alignment objective. This design allows the student to better accommodate appearance variations in aerial scenes while preserving the alignment structure induced by the teacher. Experiments on six remote sensing benchmarks indicate that AVION can improve few-shot classification performance without degrading generalization to novel categories.

Acknowledgments

This research is supported in part by grants from the National Science Foundation (OAC-2118240, HDR Institute: Imageomics).

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